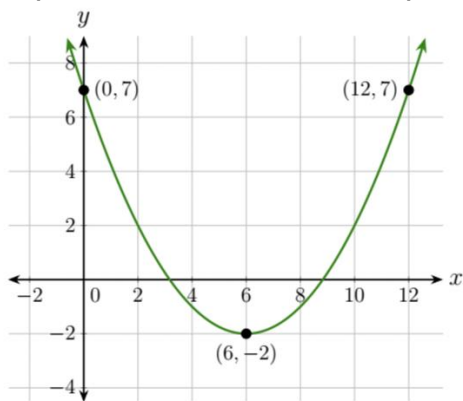


Algebra 1: Unit 9, Lesson 2

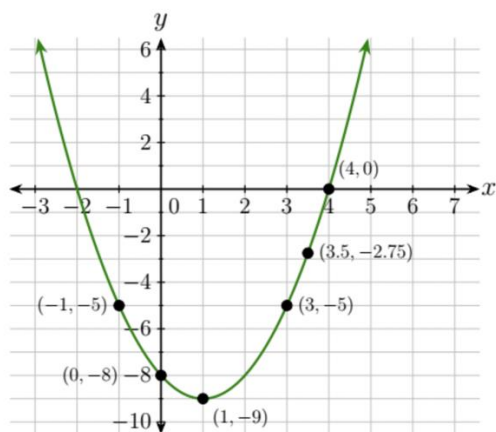
1. A quadratic function has the equation, $y = a(x + 2)^2 - 6$, and passes through the point, $(-5, 30)$. What is the value of a ?

2. A quadratic function with points, $(0, 7)$, $(6, -2)$, $(12, 7)$ is shown on the coordinate grid.



A student is writing the equation of the quadratic function in vertex form using $f(x) = a(x - h)^2 + k$. What are the values of h and k ?

3. A quadratic equation is shown on the coordinate grid. The points, $(-1, -5)$, $(0, -8)$, $(1, -9)$, $(3, -5)$, $(3.5, -2.75)$, and $(4, 0)$, are labeled on the graph.



Which pair of labeled points are equidistant from the vertex?